



PREDICTIVE MODELING FOR LENGTH OF STAY USING MACHINE LEARNING

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Introduction & Background

Introduction-

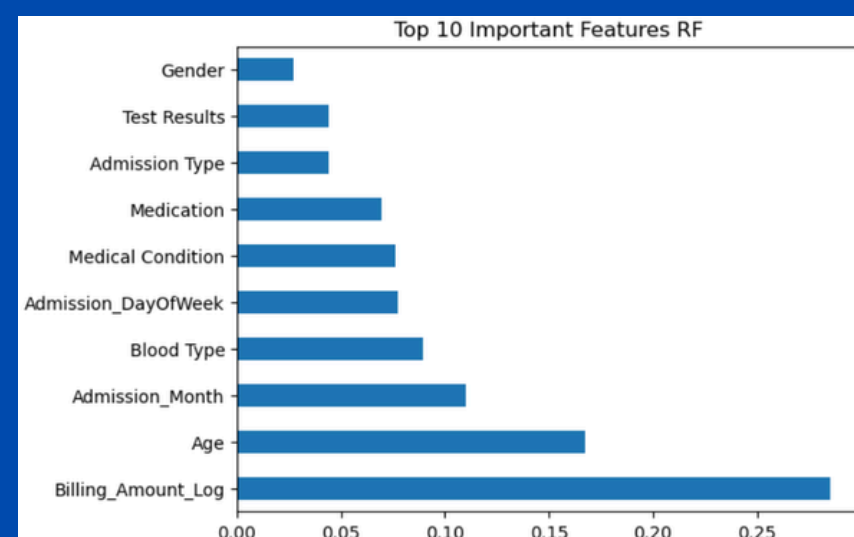
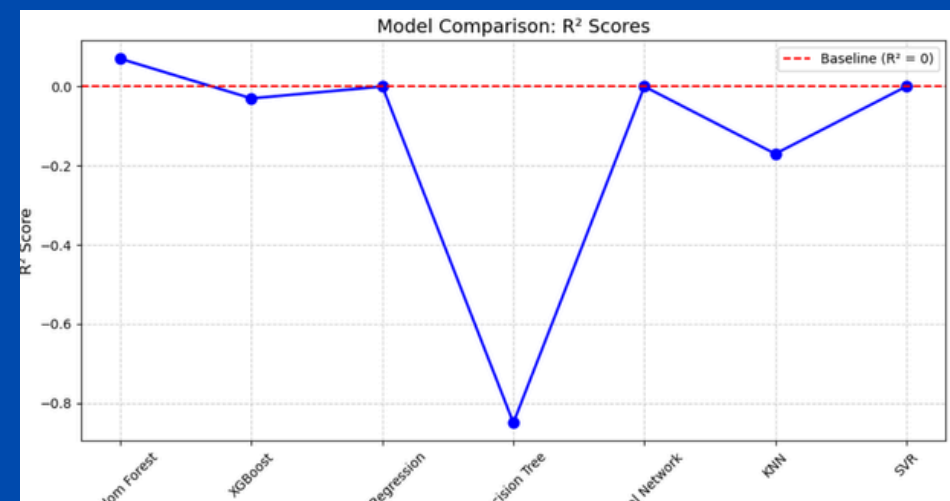
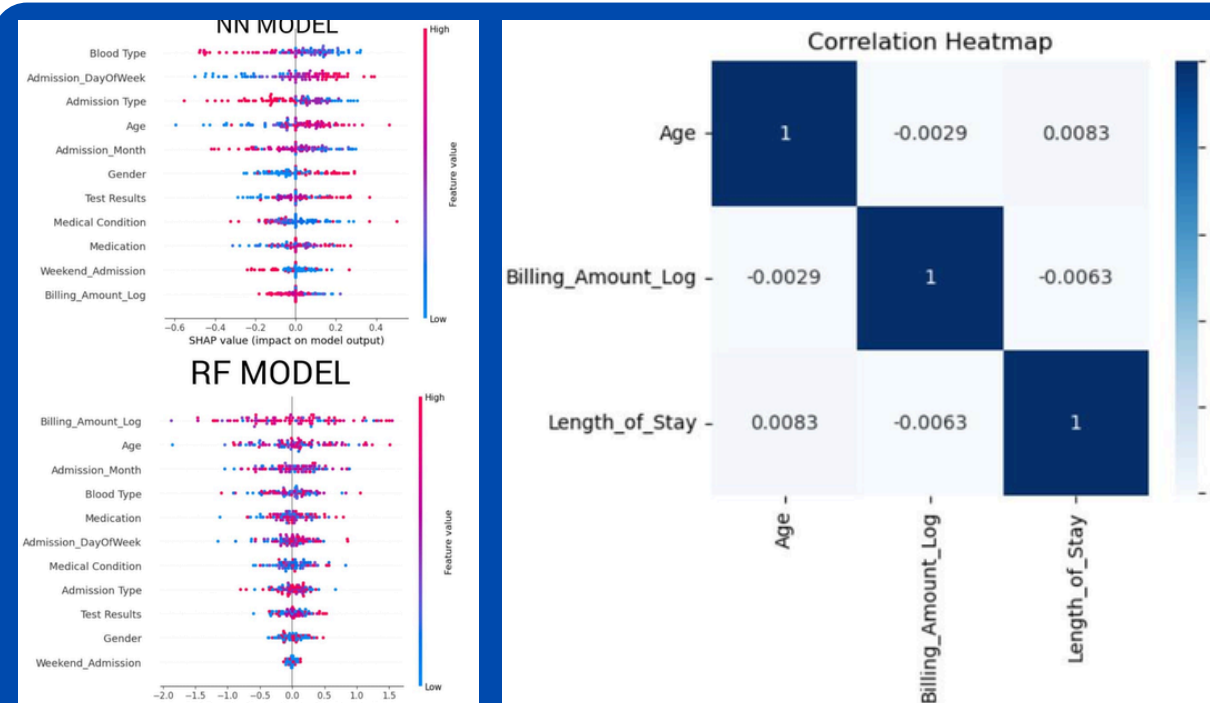
- Length of Stay (LoS) is a critical hospital performance metric [1]
- Impacts patient outcomes, discharge planning, and resource use
- Machine Learning (RF, XGBoost, NN) outperforms traditional models
- This project uses:[1]
 - Genetic Algorithms (GA) for optimal feature selection
 - SHAP for transparent, explainable predictions.

Background-

- LoS influenced by age, medical condition, admission type, billing, and season[2]
- SHAP identifies features that impact LoS predictions
- Dataset: Kaggle healthcare (demographic, clinical, billing data)
- Aim: Enhance LoS prediction using ML + GA + SHAP

Methods

- Dataset: Kaggle Healthcare – Disease Prediction (Prasad, 2025)[4]
- Included: Patients ≥ 18 yrs with complete admission/discharge data
- Excluded: Pediatric patients, incomplete or invalid records
- Preprocessing: Median/mode imputation, log transformation, standard scaling
- Encoding: Label encoding (low-cardinality), target encoding (high-cardinality)
- Engineered Features: Billing_Amount_Log, Admission_DayOfWeek, Admission_Month, interaction terms[4]
- Models: Linear Regression, Decision Tree, Random Forest, XGBoost, Neural Network, KNN, SVR[4]
- Feature Selection: Genetic Algorithm (GA) using Random Forest R^2 as fitness
- Interpretability: SHAP values & feature importance from RF and XGB
- Evaluation: 80/20 train-test split, 3-fold cross-validation
- Tools: Python (pandas, sklearn, xgboost, keras, shap, pygad, seaborn)
- Data Visualization: Used seaborn and matplotlib for EDA and model result interpretation[4]



Results

- Random Forest performed best ($R^2 \approx 0.07$, RMSE ≈ 8.36), but all models showed low predictive power. Neural Network and other models struggled due to limited and non-clinical features.
- Key predictors: Medical Condition, Medication, Billing Amount, Admission Type.
- SHAP showed complex conditions & high billing $\rightarrow$ longer LOS; admission-related factors explained shorter LOS.
- Genetic Algorithm (GA) Feature Selection
- GA identified similar key features but did not improve performance, confirming the dataset lacked richer clinical variables.
- Models were weak in prediction but useful for identifying drivers of LOS.

Conclusion

- ML models were built to predict hospital Length of Stay (LoS) using admission-time data. Key influencing features: Billing Amount, Admission Type, Medical Condition, Age, Day of Admission.[3]
- Random Forest performed best ($R^2 \approx 0.07$), but overall predictive accuracy was low.
- GA reduced feature set but did not improve model performance. Dataset lacked clinical depth needed for reliable LoS prediction. Study lays foundation for future work in LoS prediction and hospital resource optimization.[4]

Reference

- Kim, E., Lee, S., Kim, Y., et al. (2024). Hospital length of stay prediction for planned admissions using machine learning: Retrospective cohort study. *Journal of Medical Internet Research*, 26, e59260.
- Morton, A., Marquez, E., Curtis, K., et al. (2022). A systematic review of the prediction of hospital length of stay: Approaches, merits, and shortcomings. *BMC Health Services Research*, 22(1), 275. <https://doi.org/10.1186/s12913-022-07679-y>.
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